

Unsteady MHD Viscous Nanofluid Flow Containing Gyrotactic Microorganisms through a Cylindrical Outer Region

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Abstract—A conceptual boundary level experiment of heat and mass transfer flow in an MHD viscous nanofluid containing gyrotactic microorganisms around a stretching cylinder has been evaluated in this work. The governing equations of the flow have been executed by the boundary layer phenomena. The primary non-linear PDEs have been turned into a dimensionless form via application of suitable dimensionless variables. The resulting non-similar dimensionless differential equations have been resolved by the EFDM with the help of FORTRAN software version 6.6a. Stability and convergence criteria analysis enabled obtaining the convergent solutions. The consequences of numerous parameters such as the Magnetic parameter, Brownian motion parameter, Thermophoresis parameter, Lewis number, Micro-organism Lewis number, and Bio-convection parameter on the flow have been investigated, and the outcomes have been demonstrated graphically. Finally, the current outcomes were compared with the previously published results.

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INTRODUCTION

Magnetohydrodynamics (MHD) is the science involving the motion of a high conducting liquid with a magnetic effect. In 1904, Prandtl [1] established the laminar boundary layer phenomena. Faraday [2] experimented with a mercury flow in glass tubes located between magnetic poles and revealed the induced voltage through the tube due to the mercury motion. Alfven [3] established MHD waves created by disruptions. Crane [4] seemed to initiate the theory of boundary flow along a stretching ambient surface. Das et al. [5] showed the features of nanofluids, including the excess of the temperature-dependent thermal conductivity over that in the base fluid, along with a substantial increase in the thermal conductivity at low particle volume concentrations. Choi et al. [6] described the thermal conductivity enhancement for nanofluid cases. Gill et al. [7] established a model regarding free convection with a horizontal plate.

Sakiadis [8] studied the boundary layer Blasius motion from a solid surface applied with constant slit speed taking the liquid at rest. Different researchers investigated some features of such problems. Those include Hayat et al. [9] and Xu and Liao [10]. Non-uniform slot injection over a cylinder was considered by Dutta et al. [11]. Wang [12] investigated a steady two-dimensional flow of fluid across a stretching cylinder. Ishak et al. [13] studied uniform suction with heat transfer due to stretching cylinders. Ishak et al. [14] also studied an MHD flow for uniform suction and heat transfer due to stretching cylinders. Butt et al. [15] numerically investigated the effect of magnetic field on the entropy generation in a viscous flow in an expanded cylinder embedded in a porous medium. Very recently, Ferdows et al. [16] studied the heat and mass transfer in a viscous nanofluid with a gyrotactic microorganism over a stretching cylinder. Platt

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[17] used the term of bio-convection for biological nanofluid mechanics. Alloui et al. [18] investigated the heat transfer effects of gyrotactic microorganisms on a bio convection suspension. Tham et al. [19] embedded a horizontal circular cylinder in a porous medium filled with nanofluids containing gyrotactic micro-organisms.

The above writing showed that nanofluids containing gyrotactic microorganisms are attached to a cylinder with suction and demonstrates the united effect of the magnetic field. However, in this model, the nanofluid flow outside the stretching cylinder was formulated as taking a geometrical configuration. The primary equations took a non-dimensional form due to suitable dimensionless variables. The resulting non-similar dimensionless differential equations were evaluated by the EFDM. Finally, the impacts of different parameters were shown with the help of graphs.

2. MATHEMATICAL FORMULATION

A 1D unsteady laminar incompressible fluid flow of a nanofluid containing gyrotactic microorganism around a cylinder is considered. The physical configuration of the system and the coordinate model is illustrated in Fig. 1.

Here the cylinder is located along the x coordinate, and its radius is along the y coordinate. The magnetic vector B_0 is taken as $B = (0, B_0, 0)$. The cylindrical velocity $U_w = U_0(x/l)$, U_0 is taken fixed, and l is the length. The nanofluid has a constant temperature T_w and a constant concentration C_w along the stretching cylinder. The circumambient values accomplished as y tends to infinite of T and C are noted by T_∞ and C_∞ accordingly.

Under these assumptions, the governing equations are as follows.

Continuity equation

$$\frac{\partial v}{\partial y} = 0. \quad (1)$$

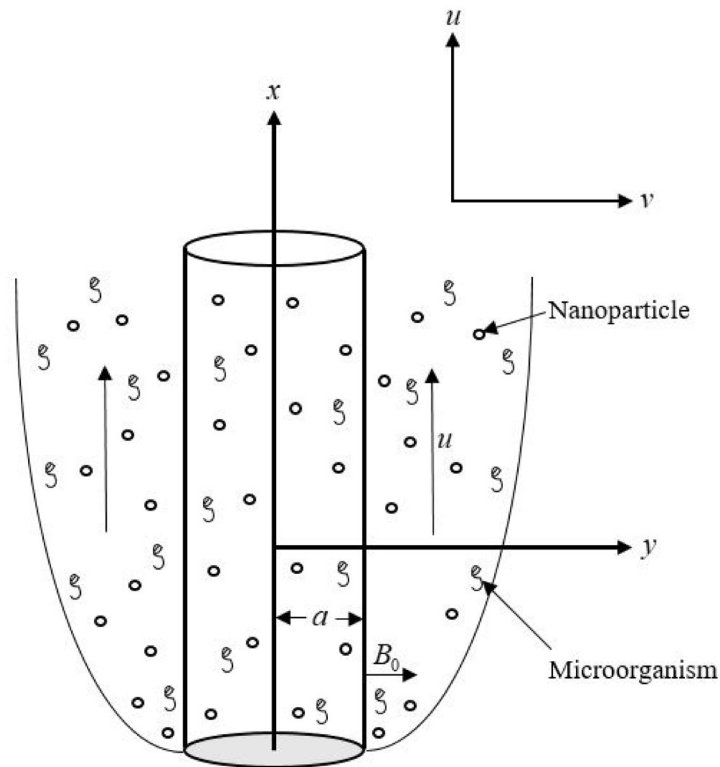


Fig. 1. Physical configuration of model.

Momentum equation

$$\frac{\partial u}{\partial t} - v_0 \frac{\partial u}{\partial y} = v \left(\frac{\partial^2 u}{\partial y^2} + \frac{1}{y} \frac{\partial u}{\partial y} \right) + g\beta(T - T_\infty) + g\beta^*(C - C_\infty) - \frac{\sigma_e B_0^2 u}{\rho}. \quad (2)$$

Energy equation

$$\frac{\partial T}{\partial t} - v_0 \frac{\partial T}{\partial y} = \alpha \left(\frac{\partial^2 T}{\partial y^2} + \frac{1}{y} \frac{\partial T}{\partial y} \right) + \frac{v}{c_p} \left(\frac{\partial u}{\partial y} \right)^2 + \tau \left(D_B \left(\frac{\partial T}{\partial y} \frac{\partial C}{\partial y} \right) + \frac{D_T}{T_\infty} \left(\frac{\partial T}{\partial y} \right)^2 \right). \quad (3)$$

Concentration equation

$$\frac{\partial C}{\partial t} - v_0 \frac{\partial C}{\partial y} = D_B \left(\frac{\partial^2 C}{\partial y^2} + \frac{1}{y} \frac{\partial C}{\partial y} \right) + \frac{v}{c_p} \left(\frac{\partial u}{\partial y} \right)^2 + \frac{D_T}{T_\infty} \frac{\partial^2 T}{\partial y^2}. \quad (4)$$

Microorganism equation

$$\frac{\partial n}{\partial t} - v_0 \frac{\partial n}{\partial y} = D_n \left(\frac{\partial^2 n}{\partial y^2} + \frac{1}{y} \frac{\partial n}{\partial y} \right) + \frac{dW_C}{C_w - C_\infty} \left\{ \frac{\partial n}{\partial y} \frac{\partial C}{\partial y} + n \frac{\partial^2 C}{\partial y^2} \right\}. \quad (5)$$

The initial and boundary conditions for the model are as follows:

$$u = U_w, \quad v = 0, \quad T = T_w, \quad C = C_w, \quad n = n_w \quad \text{at } y = a, \quad (6)$$

$$u = 0, \quad v = 0, \quad T = T_\infty, \quad C = C_\infty, \quad n = n_\infty \quad \text{as } y \rightarrow \infty, \quad (7)$$

with the dimensionless quantities,

$$Y = \frac{yU_0}{v}, \quad \tau = \frac{tU_0^2}{v}, \quad U = \frac{u}{U_0}, \quad V = \frac{v_0}{U_0}, \quad \bar{T} = \frac{T - T_\infty}{T_w - T_\infty}, \quad \bar{C} = \frac{C - C_\infty}{C_w - C_\infty}, \quad \bar{n} = \frac{n - n_\infty}{n_w - n_\infty}.$$

The resulting dimensionless differential equations are as follows:

$$\frac{\partial V}{\partial Y} = 0, \quad (8)$$

$$\frac{\partial U}{\partial \tau} - S \frac{\partial U}{\partial Y} = \frac{\partial^2 U}{\partial Y^2} + \frac{1}{Y} \frac{\partial U}{\partial Y} + G_r \bar{T} + G_m \bar{C} - MU, \quad (9)$$

$$\frac{\partial \bar{T}}{\partial \tau} - S \frac{\partial \bar{T}}{\partial Y} = \frac{1}{P_r} \frac{\partial^2 \bar{T}}{\partial Y^2} + \frac{1}{P_r Y} \frac{\partial \bar{T}}{\partial Y} + E_c \left(\frac{\partial U}{\partial Y} \right)^2 + N_b \frac{\partial \bar{T}}{\partial Y} \frac{\partial \bar{C}}{\partial Y} + N_t \left(\frac{\partial \bar{T}}{\partial Y} \right)^2, \quad (10)$$

$$\frac{\partial \bar{C}}{\partial \tau} - S \frac{\partial \bar{C}}{\partial Y} = \frac{1}{L_e} \frac{\partial^2 \bar{C}}{\partial Y^2} + \frac{1}{L_e Y} \frac{\partial \bar{C}}{\partial Y} + E_m \left(\frac{\partial U}{\partial Y} \right)^2 + S_r \frac{\partial^2 \bar{T}}{\partial Y^2}, \quad (11)$$

$$\frac{\partial \bar{n}}{\partial \tau} - S \frac{\partial \bar{n}}{\partial Y} = \frac{1}{S_{cm}} \frac{\partial^2 \bar{n}}{\partial Y^2} + \frac{1}{S_{cm} Y} \frac{\partial \bar{n}}{\partial Y} + P_e \frac{\partial \bar{n}}{\partial Y} \frac{\partial \bar{C}}{\partial Y} + P_e (\sigma + \bar{n}) \frac{\partial^2 \bar{C}}{\partial Y^2}, \quad (12)$$

and the dimensionless initial and boundary conditions can be written as follows:

$$U = 1, \quad V = 0, \quad \bar{T} = 1, \quad \bar{C} = 1, \quad \bar{n} = 1 \quad \text{at } Y = 0,$$

$$U = 0, \quad V = 0, \quad \bar{T} = 0, \quad \bar{C} = 0, \quad \bar{n} = 0 \quad \text{as } Y \rightarrow \infty. \quad (13)$$

The dimensionless parameters are as follows:

the Suction parameter $S = \frac{v_0}{U_0}$, the Grashoff number $G_r = \frac{g\beta(T_w - T_\infty)v}{U_0^3}$, the Modified Grashoff number $G_m = \frac{g\beta^*(C_w - C_\infty)v}{U_0^3}$, the Magnetic parameter $M = \frac{\sigma_e B_0^2 v}{\rho U_0^2}$, the Prandtl number $P_r = \frac{\rho C_P v}{k}$, the Eckert number $E_c = \frac{U_0^2}{C_P(T_w - T_\infty)}$, the Brownian motion parameter $N_b = \frac{\tau_{DB}(C_w - C_\infty)}{v}$, the Thermophoresis parameter $N_t = \frac{\tau_{DT}(T_w - T_\infty)}{T_\infty v}$, the Lewis number $L_e = \frac{v}{D_B}$, the Modified Eckert number $E_m = \frac{U_0^2}{C_P(C_w - C_\infty)}$, the Soret number $S_r = \frac{D_T(T_w - T_\infty)}{T_\infty v(C_w - C_\infty)}$, the Micro-organism Lewis number $S_{cm} = \frac{v}{D_n}$, the Peclet number $P_e = \frac{dW_C}{v}$, and the Bio-convection parameter $\sigma = \frac{n_w}{(n_w - n_\infty)}$.

3. SHEAR STRESS, NUSSELT NUMBER, SHERWOOD NUMBER, AND MICROORGANISM NUMBER

For the velocity profile, the local shear stress, $\tau_L = R_e \left(\frac{\partial u}{\partial y} \right)_{y=0}$ is proportional to $\left(\frac{\partial U}{\partial Y} \right)_{Y=0}$.

The Local Nusselt number $N_{uL} = (1 + R) \left(-\frac{\partial T}{\partial y} \right)_{y=0}$ is proportional to $\left(-\frac{\partial \bar{T}}{\partial Y} \right)_{Y=0}$.

The Local Sherwood number $S_{hL} = \left(-\frac{\partial C}{\partial y} \right)_{y=0}$ is proportional to $\left(-\frac{\partial \bar{C}}{\partial Y} \right)_{Y=0}$.

The Microorganism number $M_n = W_c \left(\frac{\partial n}{\partial y} \right)_{y=0}$ is proportional to $\left(\frac{\partial \bar{n}}{\partial Y} \right)_{Y=0}$.

4. CALCULATION TECHNIQUE

To figure out dimensionless primary PDEs (9)–(12) by the FDM, we consider a 2D region (Fig. 2).

Here we suppose that $Y_{MAX} (= 40)$, i.e., Y varies from 0 to 40, where $n (= 300)$ is the grid spacing and ΔY is the constant block size along the Y direction, which is taken as $\Delta Y = 0.050$ (approximately) with a thinner time variation, $\Delta \tau = 0.0005$.

Let U' , V' , $\bar{T}'$, $\bar{C}'$, and $\bar{n}'$ illustrate the data of U , V , $\bar{T}$, $\bar{C}$, and $\bar{n}$, accordingly. The FDM set of the equations is as follows:

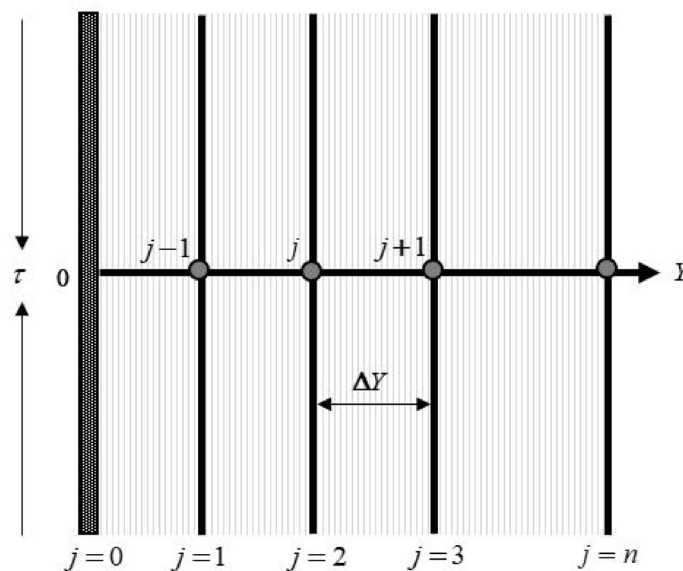


Fig. 2. The finite difference space grid.

$$\frac{V_{i,j} - V_{i,j-1}}{\Delta Y} = 0, \quad (14)$$

$$\begin{aligned} \frac{U'_{i,j} - U_{i,j}}{\Delta \tau} - S \frac{U_{i,j+1} - U_{i,j}}{\Delta Y} &= \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{(\Delta Y)^2} \\ &+ \frac{1}{Y} \frac{U_{i,j+1} - U_{i,j}}{\Delta Y} + G_r \overline{T_{i,j}} + G_m \overline{C_{i,j}} - MU_{i,j}, \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{\overline{T'_{i,j}} - \overline{T_{i,j}}}{\Delta \tau} - S \frac{\overline{T_{i,j+1}} - \overline{T_{i,j}}}{\Delta Y} &= \frac{1}{P_r} \frac{\overline{T_{i,j+1}} - 2\overline{T_{i,j}} + \overline{T_{i,j-1}}}{(\Delta Y)^2} + \frac{1}{P_r Y} \frac{\overline{T_{i,j+1}} - \overline{T_{i,j}}}{\Delta Y} \\ &+ E_c \left(\frac{U_{i,j+1} - U_{i,j}}{\Delta Y} \right)^2 + N_b \left(\frac{\overline{T_{i,j+1}} - \overline{T_{i,j}}}{\Delta Y} \right) \left(\frac{\overline{C_{i,j+1}} - \overline{C_{i,j}}}{\Delta Y} \right) + N_t \left(\frac{\overline{T_{i,j+1}} - \overline{T_{i,j}}}{\Delta Y} \right)^2, \end{aligned} \quad (16)$$

$$\begin{aligned} \frac{\overline{C'_{i,j}} - \overline{C_{i,j}}}{\Delta \tau} - \frac{\overline{C_{i,j+1}} - \overline{C_{i,j}}}{\Delta Y} &= \frac{1}{L_e} \frac{\overline{C_{i,j+1}} - 2\overline{C_{i,j}} + \overline{C_{i,j-1}}}{(\Delta Y)^2} \\ &+ \frac{1}{L_e Y} \frac{\overline{C_{i,j+1}} - \overline{C_{i,j}}}{\Delta Y} + E_m \left(\frac{U_{i,j+1} - U_{i,j}}{\Delta Y} \right)^2 + S_r \frac{\overline{T_{i,j+1}} - 2\overline{T_{i,j}} + \overline{T_{i,j-1}}}{(\Delta Y)^2}, \end{aligned} \quad (17)$$

$$\begin{aligned} \frac{\overline{n'_{i,j}} - \overline{n_{i,j}}}{\Delta \tau} - S \frac{\overline{n_{i,j+1}} - \overline{n_{i,j}}}{\Delta Y} &= \frac{1}{S_{cm}} \frac{\overline{n_{i,j+1}} - 2\overline{n_{i,j}} + \overline{n_{i,j-1}}}{(\Delta Y)^2} + \frac{1}{S_{cm} Y} \frac{\overline{n_{i,j+1}} - \overline{n_{i,j}}}{\Delta Y} \\ &+ P_e \frac{\overline{n_{i,j+1}} - \overline{n_{i,j}}}{\Delta Y} \frac{\overline{C_{i,j+1}} - \overline{C_{i,j}}}{\Delta Y} + P_e (\sigma + \overline{n_{i,j}}) \frac{\overline{C_{i,j+1}} - 2\overline{C_{i,j}} + \overline{C_{i,j-1}}}{(\Delta Y)^2}, \end{aligned} \quad (18)$$

with the boundary conditions

$$U_0^n = 1, \quad V_0^n = 0, \quad \overline{T}_0^n = 1, \quad \overline{C}_0^n = 1, \quad \overline{n}_0^n = 1 \quad \text{at } Y = 0,$$

$$U_L^n = 0, \quad V_L^n = 0, \quad \overline{T}_L^n = 0, \quad \overline{C}_L^n = 0, \quad \overline{n}_L^n = 0, \quad \text{as } Y = L \rightarrow \infty. \quad (19)$$

The stability and convergence inequities are as follows:

$$S \frac{\Delta \tau}{\Delta Y} + \frac{2}{P_r} \frac{\Delta \tau}{(\Delta Y)^2} + 2N_b \overline{C} \frac{\Delta \tau}{(\Delta Y)^2} + 2N_t \overline{T} \frac{\Delta \tau}{(\Delta Y)^2} \leq 1, \quad (20)$$

$$S \frac{\Delta \tau}{\Delta Y} + \frac{2}{L_e} \frac{\Delta \tau}{(\Delta Y)^2} \leq 1, \quad (21)$$

$$S \frac{\Delta \tau}{\Delta Y} + \frac{2}{S_{cm}} \frac{\Delta \tau}{(\Delta Y)^2} \leq 1. \quad (22)$$

With $\Delta \tau = 0.8$ and $\Delta Y = 0.2$, Eqs. (20), (21), and (22) give $E_c \ll 1$, $P_r \geq 0.375$, $S_{cm} \geq 0.01$, and $L_e \geq 0.25$.

5. RESULTS AND DISCUSSION

For illustration of the geometrical outcome of the model, the data on the velocity distribution, temperature distribution, concentration distribution, and motile microorganism were evaluated for the Brownian motion parameter, Thermophoresis parameter, Lewis number, Micro-organism Lewis number, Magnetic parameter, and Bio-convection parameter. With the help of Fortran soft. version 6.6a, the outcomes were shown graphically. For brevity, the effect of other parameters on the velocity outline, temperature distribution, concentration profile, and motile microorganism is not shown here.

5.1. Mesh Sensitivity Analysis

To make sure the outcomes of grid space for n , the calculations were performed for three various grid spaces: $n = 200$, $n = 250$, and $n = 300$. The result of primary velocity for these various values of n can be seen in Fig. 3. However, a slight change was noticed among these velocity profiles, which gives the accuracy of the taken mesh.

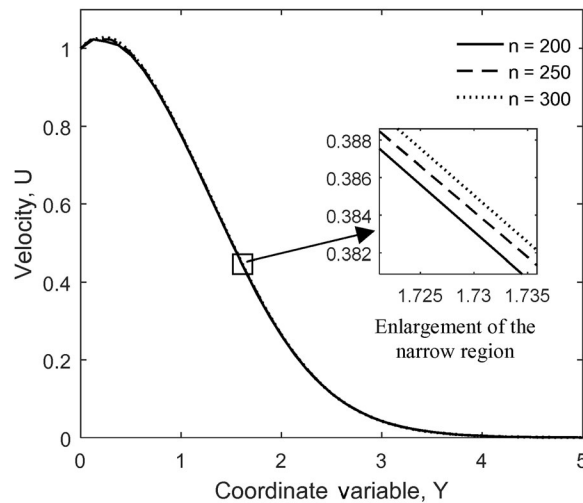


Fig. 3. Justifying grid space for mesh sensitivity test.

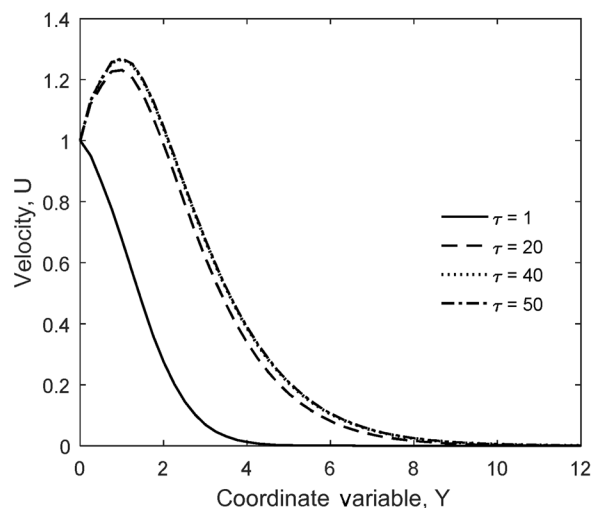


Fig. 4. Steady-state solution of velocity.

5.2. Steady-State Solution

For determination of the effects of the time step size $\Delta \tau$, the calculations were taken for four different time step sizes: $\tau = 1$, $\tau = 20$, $\tau = 40$, and $\tau = 50$. It is seen that the outcome of the problem for U , $\bar{T}$, $\bar{C}$ and $\bar{n}$ shows a slight change after $\tau = 40$. Thus, the outcomes for all variables at $\tau = 40$ are generally the steady-state solutions, which is shown in Fig. 4.

5.3. Outcome of Parameters

For determination of a clear concept of the geometrical properties of the system, the effects of various parameters (the Lewis number Le , Brownian motion parameter N_b , Thermophoresis parameter N_t , Bio-convection parameter σ , and Micro-organism Lewis number S_{cm}) on the velocity profile U , shear stress τ_L , temperature distribution T , Nusselt number distribution N_{uL} , concentration profile C , Sherwood number S_{hL} , Microorganism profile n , and Microorganism number M_n were illustrated within the boundary in Figs. 5–18.

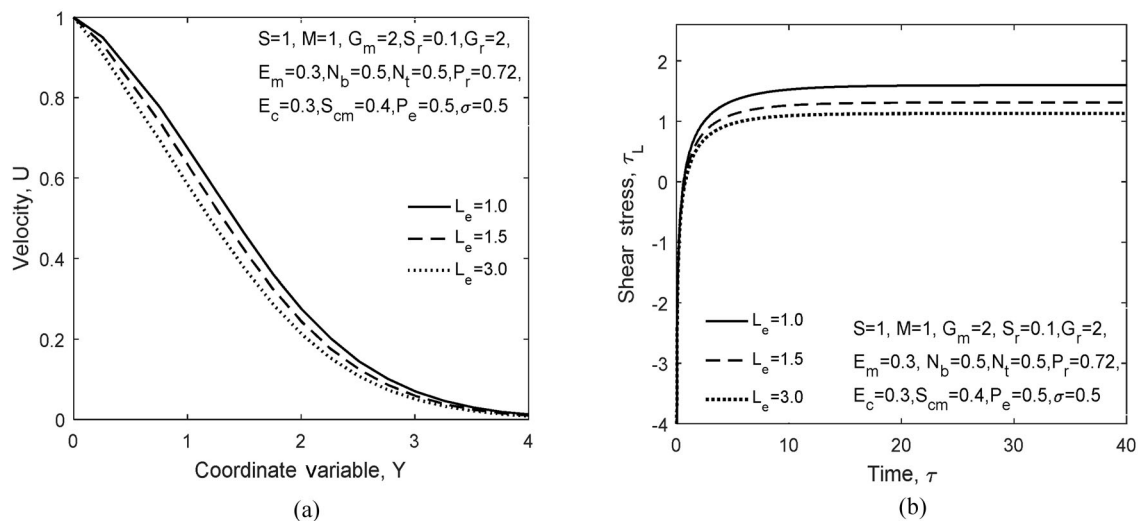


Fig. 5. (a) Velocity profile and (b) Shear stress for different values of Le .

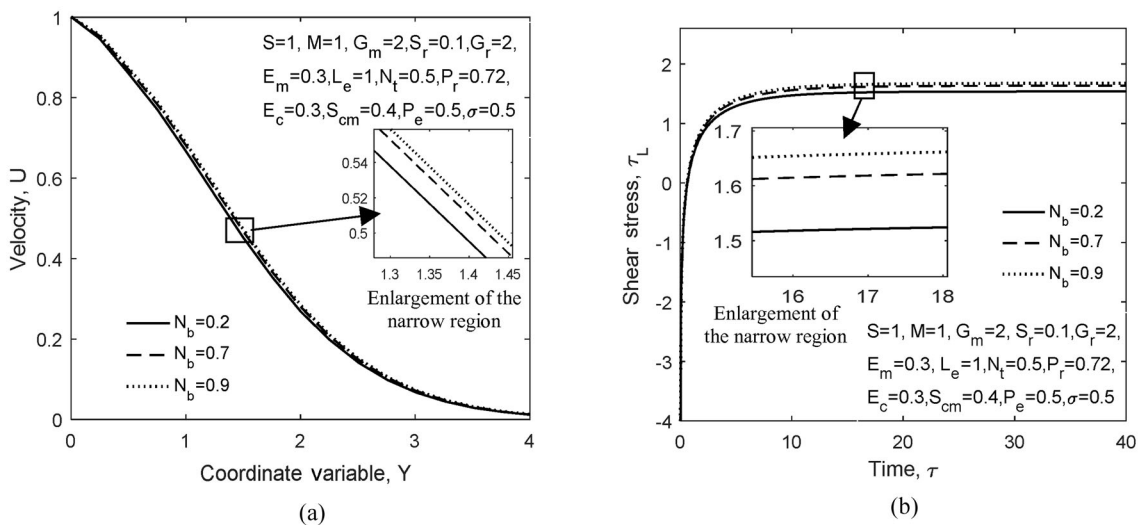


Fig. 6. (a) Velocity profile and (b) Shear stress for different values of N_b .

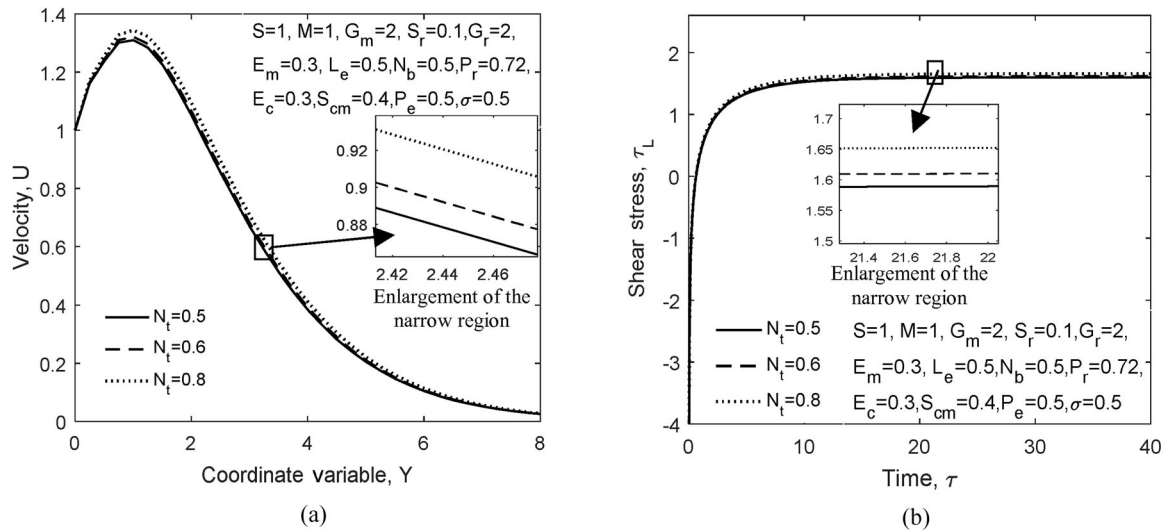


Fig. 7. (a) Velocity profile and (b) Shear stress for different values of N_t .

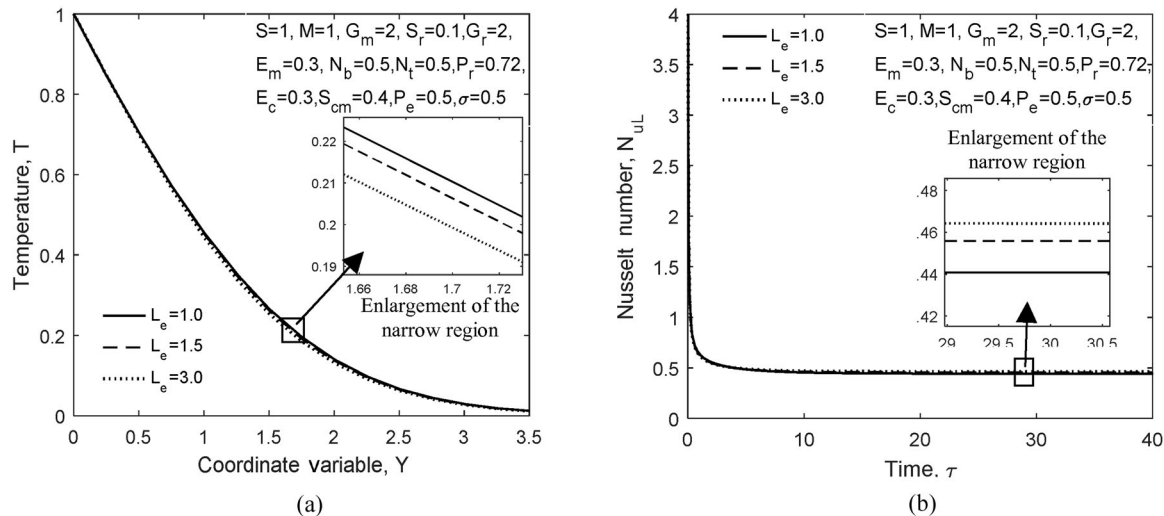


Fig. 8. (a) Temperature distribution and (b) Nusselt number distribution for different values of L_e .

Figure 5a shows the velocity profile U , and Fig. 5b displays the shear stress τ_L for different values of the Lewis number L_e . Here it is seen that the velocity profile and shear stress decrease with the increasing values of L_e .

Figure 6a shows the velocity profile U , and Fig. 6b displays the shear stress τ_L for various data of the Brownian motion parameter N_b . In the enlarged regions of these plots, it is seen that the velocity profile and shear stress grow with the increasing values of N_b .

Figure 7a shows the velocity profile U , and Fig. 7b displays the shear stress τ_L for different values of the Thermophoresis parameter N_t . In the enlarged regions of these plots it is seen that the velocity profile and shear stress grow with the increasing values of N_t .

Figure 8a shows the temperature distribution T , and Fig. 8b displays the Nusselt number distribution N_{UL} for different values of the Lewis number L_e . In the enlarged regions of these plots it is seen that the temperature distribution decreases and the Nusselt number distribution grows with the rising values of L_e .

Figure 9a shows the temperature distribution T , and Fig. 9b displays the Nusselt number distribution

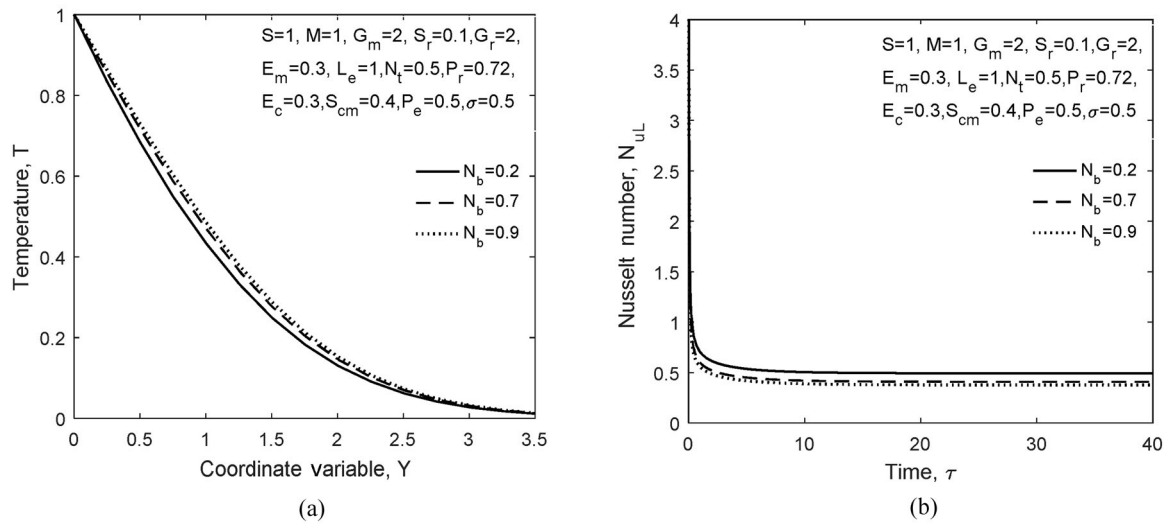


Fig. 9. (a) Temperature distribution and (b) Nusselt number distribution for different values of N_b .

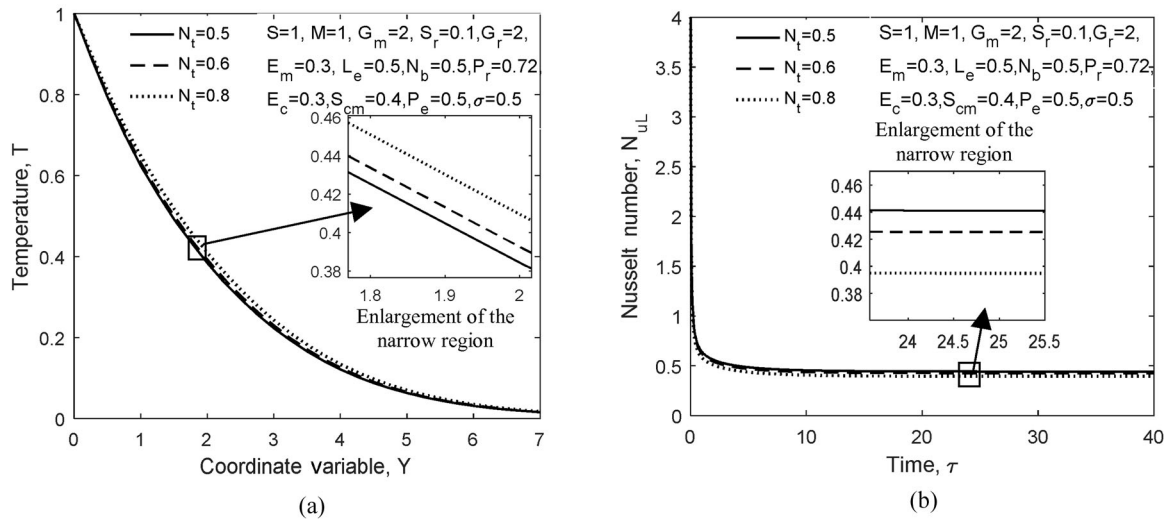


Fig. 10. (a) Temperature distribution and (b) Nusselt number distribution for different values of N_t .

N_{uL} for the values of Brownian motion parameter N_b . Here it is seen that the temperature distribution grows and the Nusselt number distribution decreases with the increasing values of N_b .

Figure 10a shows the temperature distribution T , and Fig. 10b displays the Nusselt number N_{uL} distribution for different values of the Thermophoresis parameter N_t . Here it is seen that the temperature distribution grows and the Nusselt number distribution decreases with the increasing values of N_t .

Figure 11a shows the concentration C profile, and Fig. 11b displays the Sherwood number S_{hL} for different values of the Lewis number L_e . Here it is seen that the concentration profile decreases and the Sherwood number rises with the increasing values of L_e .

Figure 12a shows the concentration C profile, and Fig. 12b displays the Sherwood number S_{hL} for various values of the Brownian motion parameter N_b . In the enlarged regions of these plots it is seen that the concentration profile rises and the Sherwood number decreases with the increasing values of N_b .

Figure 13a shows the concentration C profile, and Fig. 13b displays the Sherwood number S_{hL} for different values of the Thermophoresis parameter N_t . In the enlarged regions of these plots it is seen that the concentration profile grows and the Sherwood number decreases with the increasing values of N_t .

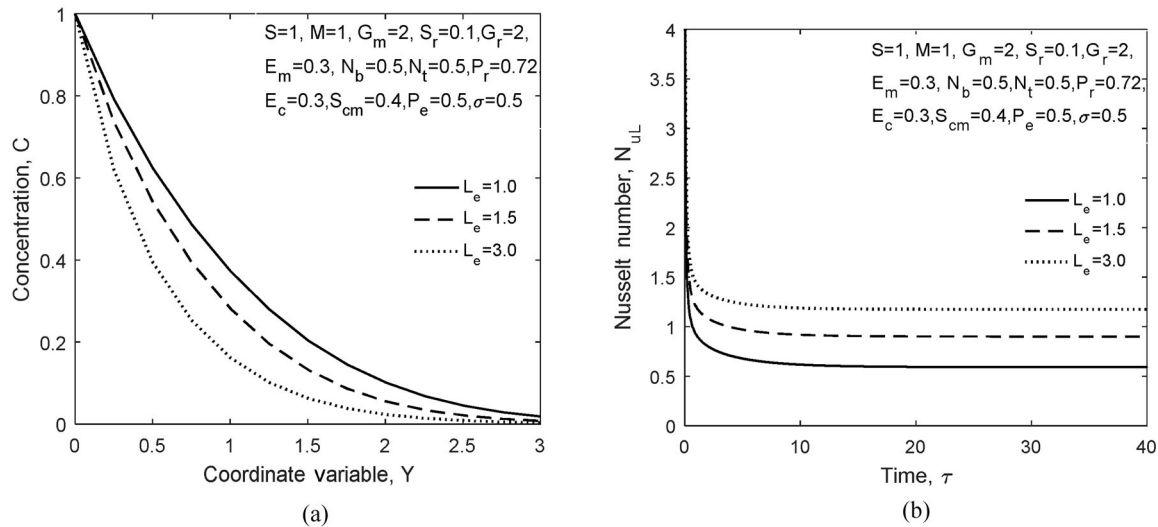


Fig. 11. (a) Temperature distribution and (b) Nusselt number distribution for different values of Le .

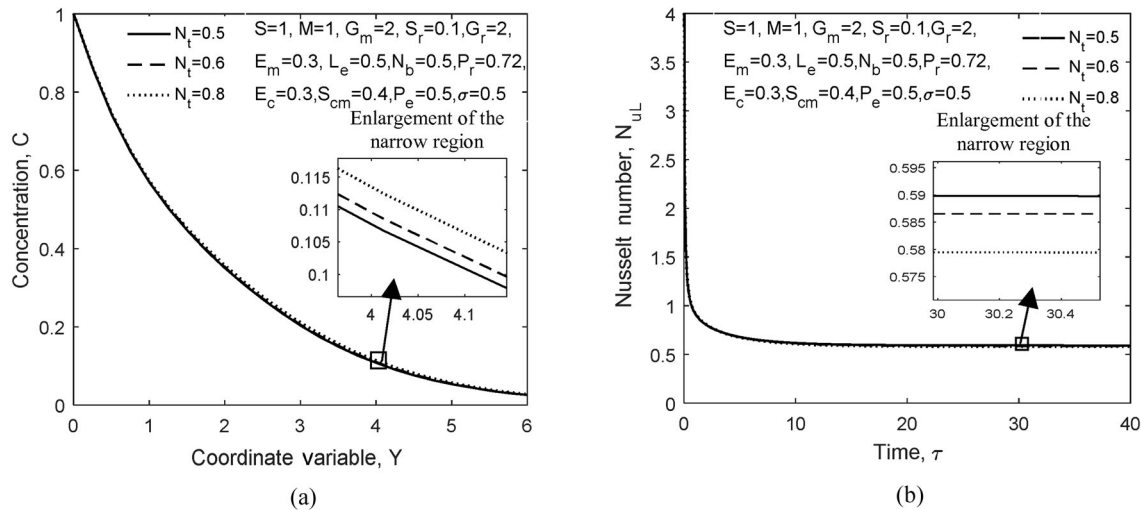


Fig. 12. (a) Concentration profile and (b) Sherwood number for different values of N_b .

Figure 14a shows the Microorganism profile n , and Fig. 14b displays the Microorganism number M_n for different values of the Lewis number Le . In the enlarged regions of these plots it is seen that the Microorganism profile and Microorganism number decrease with the rising values of Le .

Figure 15a shows the Microorganism profile n , and Fig. 15b displays the Microorganism number M_n for various values of the Brownian motion parameter N_b . In the enlarged regions of these plots it is seen that the Microorganism profile and Microorganism number decrease with the increasing values of N_b .

Figure 16a shows the Microorganism profile n , and Fig. 16b displays the Microorganism number M_n for different values of the Thermophoresis parameter N_t . In the enlarged regions of these plots it is seen that the Microorganism profile and Microorganism number decrease with the increase in N_t .

Figure 17a shows the Microorganism profile n , and Fig. 17b displays the Microorganism number M_n for different values of the Bio-convection parameter σ . In the enlarged regions of these plots it is seen that the Microorganism profile and Microorganism number increase with the increasing values of σ .

Figure 18a shows the Microorganism profile n , and Fig. 18b displays the Microorganism number M_n for various data of the Microorganism Lewis number S_{cm} . Here it is seen that the Microorganism profile and Microorganism number decrease with the increasing values of S_{cm} .

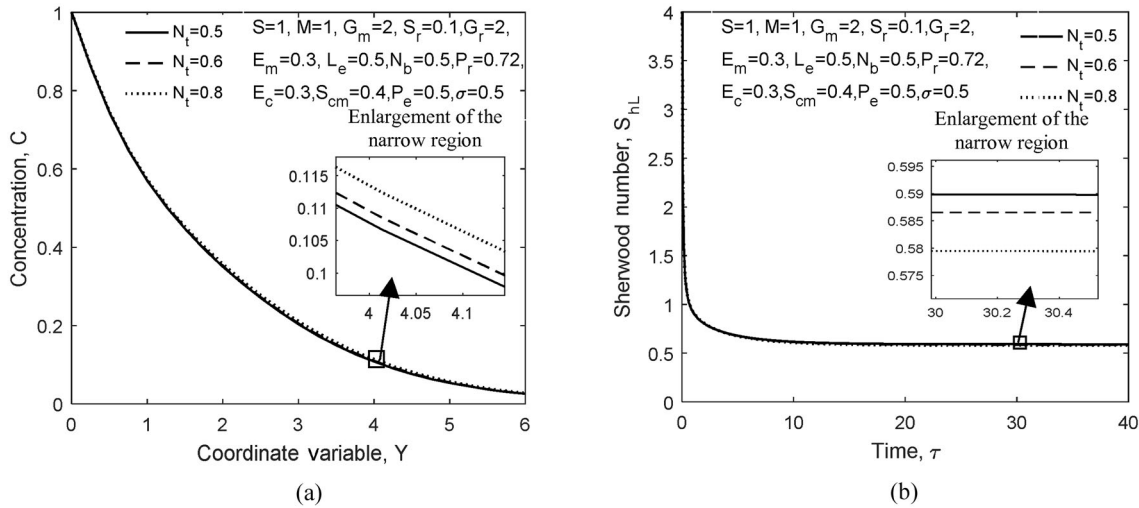


Fig. 13. (a) Concentration profile and (b) Sherwood number for different values of N_t .

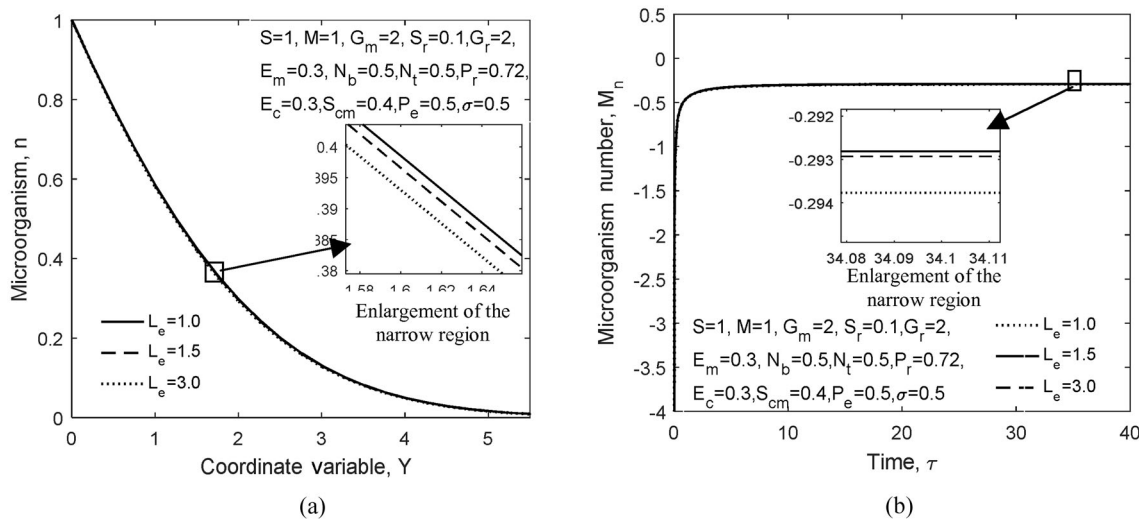


Fig. 14. (a) Microorganism profile and (b) Microorganism number at the wall for different values of Le .

5.4. Comparison

Finally, two qualitative comparisons between the present results and the published results of Ferdows et al. [16] are displayed in Figs. 19 and 20.

Thus, the obtained results are qualitatively the same as the results of Ferdows et al. [16], yet differing quantitatively. So, both systems have shown the same kinds of results, as displayed in Figs. 19 and 20, but the results of the present investigation are not exactly quantitatively the same as the results of Ferdows et al. [16].

CONCLUSIONS

The EFDM solution for an unsteady MHD viscous nanofluid flow containing gyrotactic microorganisms over a cylindrical region has been investigated. Some findings of this investigation are listed below.

- The velocity profile increases for the rising values of N_b and N_t and decreases for the rising values of M and Le .

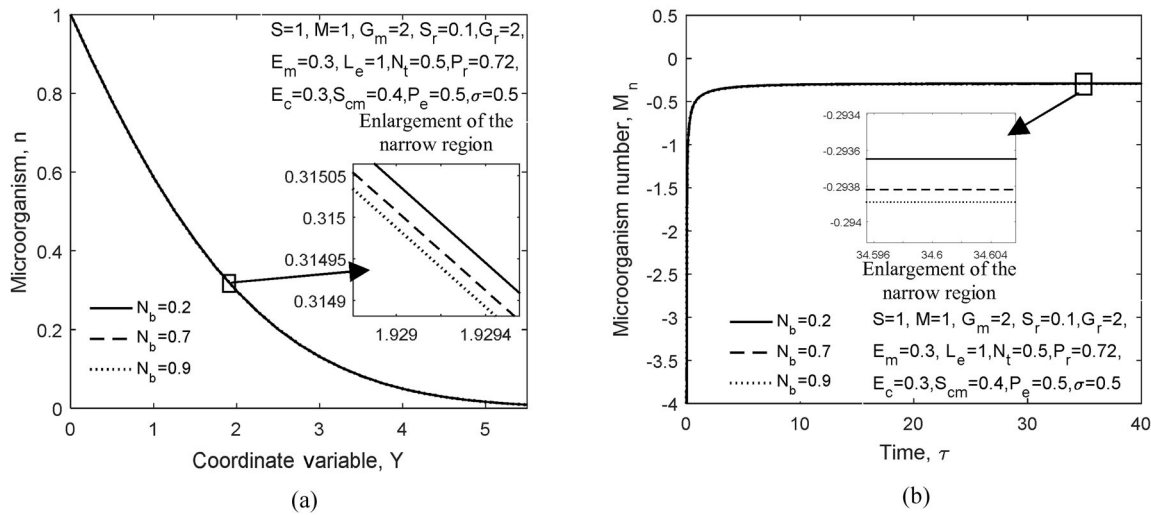


Fig. 15. (a) Microorganism profile and (b) Microorganism number at the wall for different values of N_b .

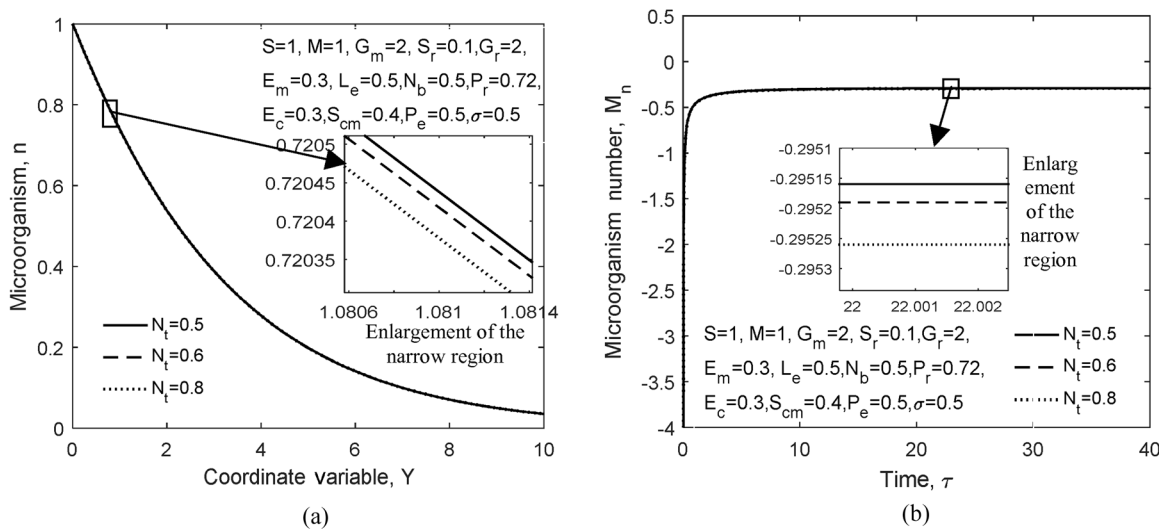


Fig. 16. (a) Microorganism profile and (b) Microorganism number at the wall for different values of N_t .

- The shear stress increases for the rising values of N_b and N_t and decreases for the rising values of L_e .
- The temperature distribution increases for the rising values of N_b and N_t and decreases for the rising values of L_e and P_r .
- The Nusselt number distribution increases for the rising values of L_e and decreases for the rising values of N_b and N_t .
- The concentration profile increases for the rising values of N_b and N_t and decreases for the rising values of L_e .
- The Sherwood number increases for the rising values of L_e and decreases for the rising values of N_b and N_t .
- The microorganism profile increases for the rising values of σ and decreases for the rising values of N_b , N_t , L_e , and S_{cm} .
- The microorganism number at the wall increases for the rising values of σ and decreases for the rising values of N_b , N_t , L_e , and S_{cm} .

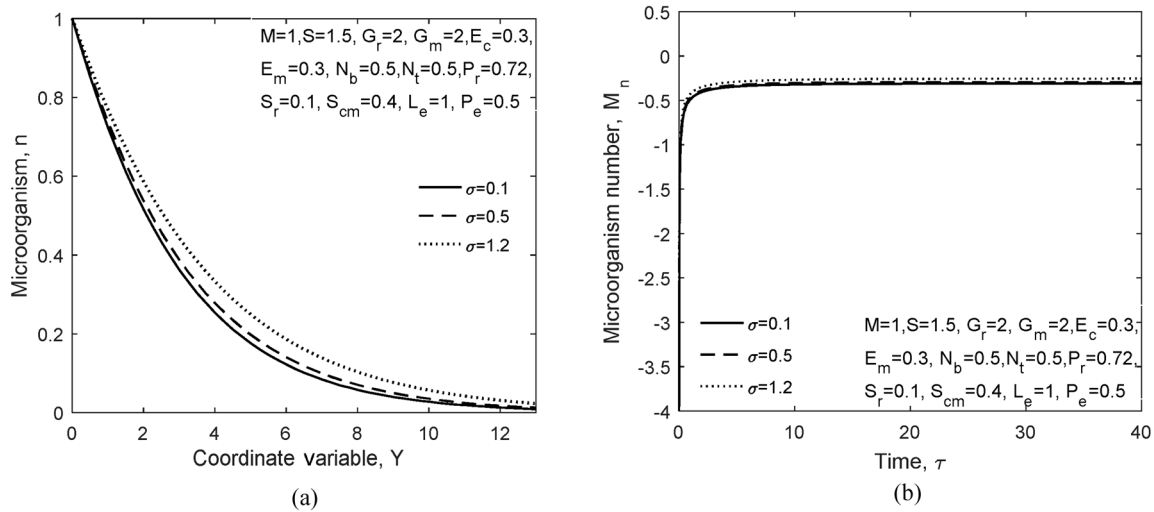


Fig. 17. (a) Microorganism profile and (b) Microorganism number at the wall for different values of σ .

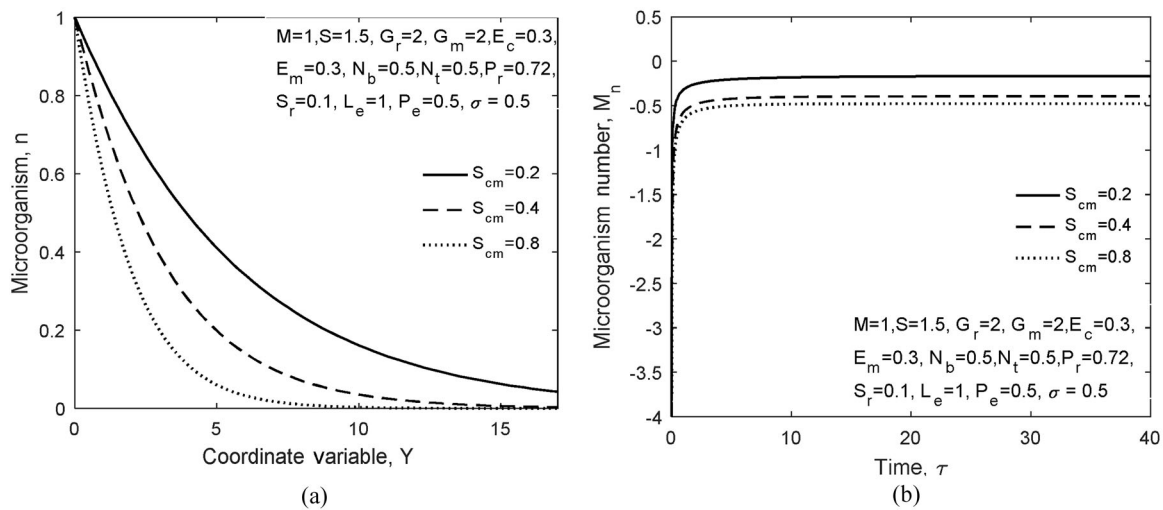


Fig. 18. (a) Microorganism profile and (b) Microorganism number at the wall for different values of S_{cm} .

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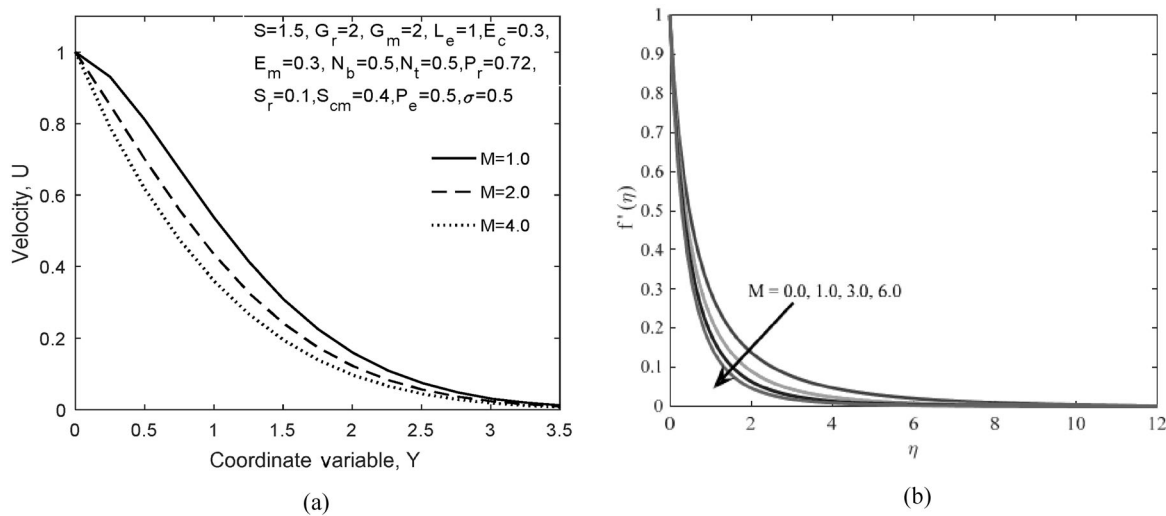


Fig. 19. Effect of Magnetic parameter (M) on (a) velocity distribution of the present study and (b) velocity distribution of Ferdows et al. [16].

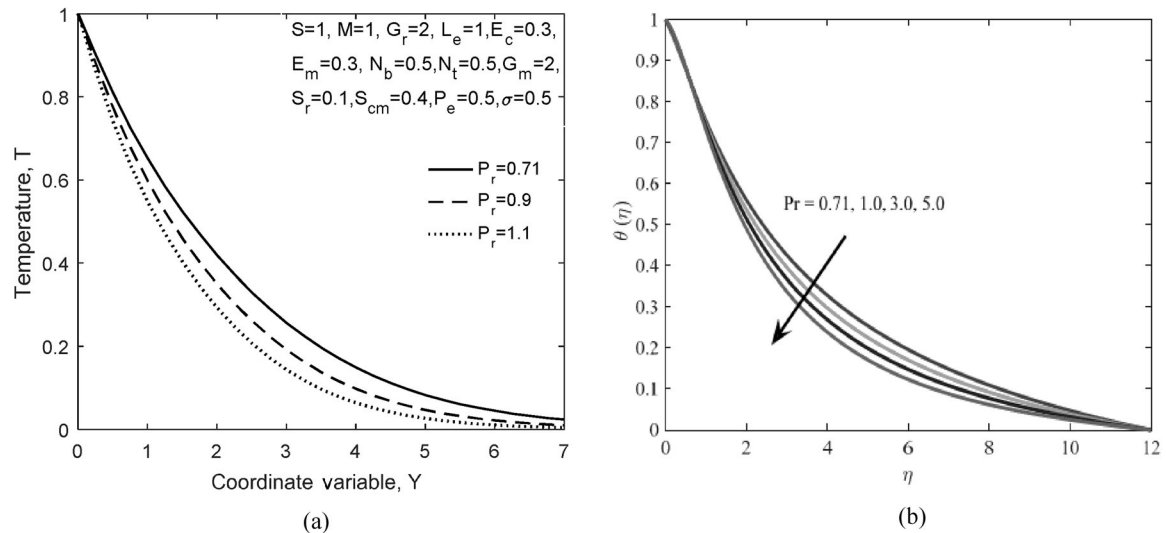


Fig. 20. Effect of Lewis number (L_e) on (a) concentration profile of the present study and (b) concentration profile of Ferdows set al. [16].

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